From Simulation to Real Robots: Practical Adaptation of Classical MAPF Algorithms for Multi-Robot Transportation

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***Abstract*—Multi-Agent Path Finding (MAPF) algorithms perform exceptionally well in simulation, where they compute collision-free paths under ideal conditions. However, deploying these algorithms on real robots introduces practical challenges, such as synchronization issues, dynamic environment changes, and communication delays, that classical MAPF models rarely address. In this paper, we present an execution-aware adaptation of two well-known MAPF algorithms: Improved Conflict-Based Search (ICBS) and Rolling Horizon Collision Resolution (RHCR), targeting multi-robot transportation tasks. Instead of developing a brand-new planner, we extend these existing methods with practical mechanisms to ensure reliable execution on physical platforms. Our adaptations include segmented planning and execution synchronization for ICBS, dynamic occupancy updates for both planners, and a modular integration with ROS2. This framework combines high-level task assignment using a Vehicle Routing Problem with Pickup and Delivery (VRPPD) formulation, path planning, and autonomous execution on physical DOTS mobile robots. We validated our approach through experiments in both Gazebo and with real robots. The results show that our modifications maintain efficient planning times and ensure reliable execution under real-world conditions, bridging the gap between theoretical MAPF models and practical logistics applications.**

***Index Terms*—multi-agent path finding, multi-robot systems, autonomous transportation, ROS2**

I. Introduction

Autonomous transportation has become an important research topic in robotics due to its applications in warehouse automation, industrial logistics, and smart manufacturing. In these environments, multiple mobile robots must transport goods efficiently while sharing the same workspace and avoiding collisions, making coordination increasingly challenging as the number of robots grows.

Multi-Agent Path Finding (MAPF) is one of the most common formulations for multi-robot coordination. Given a set of agents, their initial positions, and their target locations, MAPF algorithms compute collision-free paths while optimizing objectives such as makespan or flowtime. Algorithms such as Conflict-Based Search (CBS), Improved Conflict-Based Search (ICBS), and Rolling Horizon Collision Resolution (RHCR) offer different trade-offs between solution quality, computational cost, and scalability. However, their deployment in real transportation systems remains challenging due to assumptions of static environments, accurate trajectory execution, and reliable communication, which may be violated by localization errors, communication delays, and occupancy changes during pickup-and-delivery operations.

In this work, we adapt ICBS and RHCR for autonomous multi-robot transportation by introducing practical mechanisms for physical robot deployment. Our framework combines high-level task allocation using a Vehicle Routing Problem with Pickup and Delivery (VRPPD) formulation, MAPF-based path planning, and ROS2-based control on physical DOTS robots [1]. The proposed execution-aware adaptations include segmented planning and execution synchronization for ICBS, dynamic occupancy updates for both planners, and modular ROS2 integration to support reliable execution on physical robotic platforms.

The main contributions of this work are:

- An execution-aware adaptation of both ICBS and RHCR to support multi-robot pickup-and-delivery logistics.
- A fully integrated framework combining task allocation, path planning, ROS2 communication, and coordinated execution on a physical robotic platform.

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- Experimental validation in both simulation and real-world scenarios, demonstrating that theoretical MAPF algorithms can be successfully transferred to physical transportation tasks.

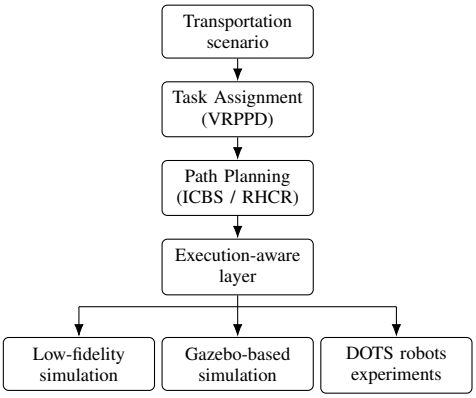


Fig. 1: Overview of the proposed transportation framework. A transportation scenario is first solved by the VRPPD task assignment module, followed by MAPF-based path planning. The resulting plans are executed by the execution layer and evaluated through low-fidelity simulation, Gazebo simulation, and experiments with real DOTS robots.

We validate our approach through Gazebo simulations across different scenario densities, evaluating its performance under varied conditions. A real-world experiment with three DOTS mobile robots is then used as a proof-of-concept to demonstrate the integration and practical feasibility of the proposed framework. The implementation is publicly available at the following repository, which also includes demonstration videos of the simulations and real-robot experiments: https://github.com/carolinaasantos/bepe-research-internship.

II. Related Work

Multi-Agent Path Finding (MAPF) serves as a foundational paradigm for coordinating teams of autonomous agents in shared workspaces, optimizing metrics such as flowtime or makespan while ensuring collision-free navigation [2]. As warehouse automation and smart manufacturing expand, traditional single-shot formulations have evolved into lifelong variants, where agents are continuously assigned new targets after completing a task [3]. However, a key challenge in modern robotics is bridging the gap between theoretical grid-based simulation and the practical constraints of deploying planners on physical multi-robot systems.

Among search-based MAPF solvers, Conflict-Based Search (CBS) stands out as an influential, optimal two-level hierarchical framework [4]. To mitigate its combinatorial complexity in dense environments, Boyarski et al. introduced Improved CBS (ICBS), incorporating enhancements such as conflict prioritization and node bypass techniques [5]. While ICBS is a state-of-the-art optimal baseline, it fundamentally assumes a static environment where complete paths are generated before execution. This limitation degrades its performance in pickup-and-delivery scenarios, where physical asset manipulation dynamically alters map topology and cell occupancy.

To handle continuous task streams at scale, the Rolling-Horizon Collision Resolution (RHCR) framework decomposes the lifelong problem into a sequence of windowed MAPF instances [6]. RHCR executes a windowed solver at regular intervals, resolving collisions within a bounded planning horizon. While this approach provides excellent scalability, the baseline algorithm implicitly assumes a static workspace layout between planning windows. In real-world multi-robot transportation tasks, object manipulation dynamically changes the environmental state. Thus, rolling horizon frameworks must be extended with execution-level feedback to adapt their environmental representations.

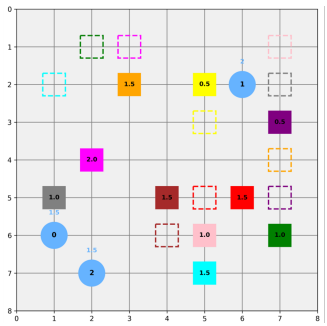
Translating these theoretical planners to real-world multi-robot systems requires addressing idealized assumptions about zero-latency communication and precise kinematics. Recent efforts have integrated kinodynamic constraints into path generation to convert discrete grid steps into smooth trajectories for physical platforms [7], while others have integrated high-level task allocation, often modeled as a Vehicle Routing Problem with Pickup and Delivery (VRPPD), with prioritized planning [8]. More recently, realistic experimental platforms have been proposed to evaluate MAPF algorithms under real-world conditions, providing scalable testbeds that bridge simulation and physical deployment [9]. Modern middleware, particularly ROS2, has also been adopted to orchestrate asynchronous navigation tasks [10]. Rather than proposing a novel planning heuristic, this work addresses this research gap by focusing on the practical system architecture required to adapt existing state-of-the-art optimal (ICBS) [5] and bounded lifelong (RHCR) [6] planners through generic execution-aware layers, specifically dynamic occupancy feedback, segmented task execution, and ROS2-centric synchronization, enabling robust deployment in physical logistics systems. Unlike previous works that modify the planner itself or incorporate kinematic constraints, our contribution focuses on execution-aware adaptations while preserving the original planners.

III. Proposed Framework

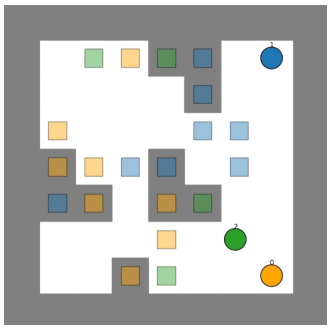
A. System Overview

The proposed framework integrates three independent modules: task allocation, MAPF planning, and robot execution. In the considered warehouse transportation scenario, a set of DOTS robots receives pickup-and-delivery requests, where each request consists of transporting a box from a pickup location to a designated delivery location. Transportation requests are first assigned through a VRPPD solver, solved using the Google OR-Tools routing solver [11], producing pickup-and-delivery sequences for each robot. ICBS or RHCR then computes collision-free trajectories, incorporating the proposed execution-aware adaptations. Finally, trajectories are executed through a ROS2-based architecture that manages communication, localization, synchronization, and robot control. Communication between the central computer and the robots is

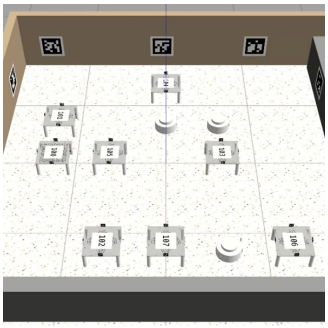
handled through ROS2 over the wireless network. The central computer transmits planned trajectories to each robot, while the robots report their estimated poses, execution status, and task-completion events. Robots do not communicate directly with one another; localization and low-level motion control are performed onboard. This modular design enables the same planning framework to operate in both simulation and physical DOTS robots. Fig. 2 illustrates the overall system architecture.



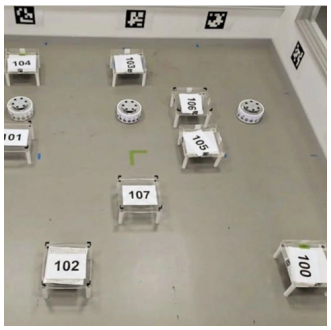
(a) Robots and boxes in the transportation scenario.



(b) Low-fidelity simulation of MAPF planners.



(c) High-fidelity simulation in Gazebo.



(d) Real experiments using DOTS robots.

Fig. 2: Overview of the transportation environment used in the proposed framework. The real warehouse scenario is represented as a grid for task assignment and MAPF planning. The resulting plans are evaluated in a lightweight Python-based low-fidelity trajectory visualizer, high-fidelity Gazebo simulations and experiments with real DOTS robots.

B. Execution-Aware MAPF Adaptations

Classical MAPF algorithms are commonly designed under assumptions that simplify planning and execution, including static environments, fixed goals, and perfect trajectory tracking. While these assumptions facilitate algorithmic development, they limit the applicability of these methods to autonomous transportation systems with physical robots.

The transportation scenario addressed in this work introduces practical requirements not explicitly represented in standard MAPF formulations. Robots must execute pickup-and-delivery operations, manipulate movable objects, synchronize transportation stages, and maintain accurate localization while coordinating with communication and motion control modules. Furthermore, environmental occupancy changes dynamically as objects are collected and delivered. The proposed solution builds upon ICBS [5] and RHCR [6], introducing execution-aware adaptations to enable reliable deployment in autonomous transportation scenarios.

To bridge the gap between theoretical planning and practical deployment, an execution layer was developed for the selected MAPF algorithms. Some adaptations are shared by both paradigms, such as dynamic occupancy updates and ROS2 integration, while others are algorithm-specific. ICBS was modified to support segmented planning throughout transportation missions and execution synchronization, whereas RHCR was extended to incorporate dynamic replanning compatible with changing environment occupancy.

The following subsections describe these adaptations and discuss their role in enabling reliable execution on real robotic platforms.

1) Dynamic Occupancy Adaptation: Classical MAPF formulations typically assume that the environment remains unchanged throughout planning. In autonomous transportation systems, however, robots continuously manipulate movable objects, causing the occupancy map to evolve during mission execution. To address this limitation, both ICBS and RHCR were extended with a dynamic occupancy mechanism that updates the planning environment whenever transportation tasks are completed. As robots collect or deliver objects, the corresponding cells are inserted into or removed from the occupancy map before subsequent planning stages. In ICBS, occupancy updates occur between consecutive planning segments, ensuring that each phase reflects the current environment. In RHCR, occupancy information is integrated into the rolling-horizon replanning process, allowing obstacle updates to propagate naturally across planning windows. This adaptation enables both algorithms to operate in transportation environments with evolving obstacle configurations, improving consistency between planning and execution.

2) Segmented Planning for ICBS: ICBS computes collision-free trajectories over the complete planning horizon, assuming the environment remains unchanged throughout execution. Since this assumption is incompatible with transportation tasks, where robots continuously collect and deliver objects, the original ICBS formulation was extended with a segmented planning strategy. The transportation mission is divided into consecutive planning segments, and after each segment is executed, the environment representation is updated according to the current state of transported objects before generating the next planning instance. This approach preserves the conflict-resolution capabilities of ICBS while allowing the planner to adapt to environment changes without modifying the underlying conflict-based search mechanism.

3) Dynamic Replanning in RHCR: Unlike ICBS, RHCR naturally performs repeated planning through a rolling-horizon strategy. Rather than computing complete trajectories for the entire mission, the planner periodically generates new paths over limited planning windows while robots continue executing previously computed actions. This makes RHCR particularly suitable for environments with changing occupancy. However, the original implementation assumes that the

environment remains fixed between replanning iterations. To support transportation tasks, the replanning process was extended to incorporate occupancy updates resulting from object manipulation. Whenever robots complete pickup or delivery operations, the environment representation is updated before the next planning horizon is generated. By integrating these updates into the rolling-horizon framework, RHCR can adapt its plans to the current transportation state while preserving its scalability and responsiveness in real multi-robot scenarios.

4) Execution Synchronization: The segmented planning strategy adopted for ICBS requires execution synchronization, since each planning segment assumes a fixed environment. After completing a segment, robots report task completion through the ROS2 communication layer, and replanning begins only after all robots finish the current stage. This synchronization ensures that occupancy updates resulting from pickup and delivery operations are consistently applied before generating the next planning segment, maintaining consistency between the planning model and the actual workspace state. In contrast, RHCR requires no explicit synchronization, as occupancy changes are naturally incorporated during successive rolling-horizon replanning cycles.

5) ROS2 Integration: The planning algorithms were integrated into a ROS2-based software architecture that converts the grid-based trajectories generated by the MAPF planners into continuous navigation waypoints for the DOTS robots. Trajectory commands are transmitted through ROS2 topics to the robot controllers, which handle navigation, motion execution, and task progression while monitoring robot status for planning synchronization. The robots also report their estimated poses, execution status, and task-completion events to the central computer, enabling coordination between planning and execution. The low-level controllers generate linear and angular velocity commands to track the waypoints and ensure smooth path following, while ROS2 provides communication between software components. This modular architecture decouples planning from robot control, facilitating deployment in both simulation and physical robotic platforms.

6) Real-World Constraints: Deploying MAPF algorithms on physical robots introduces practical challenges typically neglected in theoretical formulations. Unlike simulation environments, physical robots are affected by localization uncertainty, communication latency, actuator inaccuracies, and deviations from planned trajectories. These factors directly influence coordinated transportation missions and must be considered during system integration. The proposed execution-aware framework accommodates these limitations while preserving the planning behavior of the original algorithms. Onboard localization continuously estimates each robot's position using ArUco markers, and the resulting pose estimates are transmitted to the ROS2 framework for trajectory tracking and mission monitoring. Communication through ROS2 enables coordinated interaction between planning and execution modules, while synchronization mechanisms reduce inconsistencies caused by asynchronous task completion. Rather than modifying the optimization objectives of ICBS or RHCR, the proposed adaptations focus on improving their robustness during physical execution. This preserves the original planning characteristics of both algorithms while extending their applicability to autonomous transportation systems under realistic execution conditions.

IV. Experimental Setup

The implemented framework was evaluated through simulation and real-world experiments. Simulation was used to assess planning performance under controlled conditions and compare the adapted ICBS and RHCR algorithms across transportation scenarios of different complexity. Physical experiments then validated the complete execution pipeline on autonomous robots in a real environment.

A. Simulation Environment

Simulation experiments were conducted using Gazebo integrated with ROS2. The simulated environment reproduces the layout of the Swarm Arena at the Bristol Robotics Laboratory, including robot dimensions, transportation objects, obstacle locations, and navigation constraints. The environment is represented as a grid graph, where each vertex corresponds to a valid robot or object position, matching the MAPF formulation adopted in this work.

To evaluate the framework under different operating conditions, 36 transportation scenarios were generated with increasing complexity: 12 scenarios with 8 boxes, 12 with 9 boxes, and 12 with 10 boxes. In each scenario, three robots with different carrying capacities, together with boxes of varying weights, were randomly assigned initial positions, while each box was associated with a predefined delivery location. After task allocation through the VRPPD solver, either ICBS or RHCR was used to compute collision-free trajectories considering the execution-aware adaptations described in the previous section. The randomized scenarios allowed the evaluation of the proposed framework across different initial configurations while maintaining consistent experimental conditions. For each scenario, planning time, makespan, and task completion were recorded for performance analysis.

To ensure reproducibility, ICBS was configured with standard/disjoint splitting and segmented planning through VRPPD paths, using ICBS/CBS as the high-level solver. RHCR used Priority-Based Search (PBS) as the high-level solver, Safe Interval Path Planning (SIPP) as the low-level solver, a planning window of 20 timesteps, and a simulation window of 5.

B. Physical Experimental Setup

To evaluate the proposed framework under real operating conditions, experiments were conducted using the DOTS (Distributed Organization and Transportation System) [1] robotic platform at the Bristol Robotics Laboratory. The DOTS are low-cost omnidirectional mobile platforms designed for autonomous logistics tasks, featuring onboard computing, wireless communication, multiple sensing capabilities (including cameras and laser scanners), and a lifting mechanism for

pickup-and-delivery operations. Robot localization was performed onboard using ArUco markers deployed in the Swarm Arena, $3.7m \times 3.7m$, and the estimated poses were continuously communicated to the ROS2 framework for navigation and mission monitoring.

The complete transportation pipeline, including task allocation, MAPF planning, ROS2 communication, and trajectory execution, was deployed without modifying the planning algorithms. Three DOTS robots executed coordinated transportation missions while sharing the same workspace. The main objective was to evaluate whether the proposed adaptations enabled reliable deployment under real operating conditions.

V. Results and Discussion

A. Simulation Results

The execution-aware adaptations were evaluated in simulation using transportation scenarios with increasing levels of complexity. In each scenario, both algorithms generated collision-free trajectories while supporting the execution requirements of pickup-and-delivery operations.

TABLE I: Comparison of solutions found by both algorithms.

Algorithm	8 boxes	9 boxes	10 boxes
ICBS	12/12 (100%)	10/12 (83.33%)	9/12 (75%)
RHCR	12/12 (100%)	10/12 (83.33%)	9/12 (75%)

Number of scenarios successfully resolved.

As shown in Table I, both ICBS and RHCR solved all scenarios with 8 boxes, while the success rate decreased to 75% with 10 boxes. This reduction was mainly caused by two factors. In some cases, delivery locations became blocked by boxes not yet collected, making the scenario impossible to complete. In the remaining cases, ICBS exceeded the planning time limit as the search became more computationally expensive in denser environments, whereas RHCR exceeded the maximum mission time because repeated replanning occasionally left the robots stuck in local deadlocks.

TABLE II: Comparison of average planning time.

Algorithm	8 boxes	9 boxes	10 boxes
ICBS	17.04 ± 11.99	18.46 ± 10.95	16.38 ± 5.34
RHCR	20.24 ± 5.48	25.05 ± 10.35	35.69 ± 15.27

[a]Results are presented as mean $\pm$ standard deviation (ms).

As shown in Table II, RHCR's average planning time increased as the number of boxes grew, reaching 35.69 ± 15.27 ms in the most complex scenarios, while segmented ICBS remained relatively stable at approximately 17–18 ms. This difference is mainly due to their planning strategies: segmented ICBS plans one pickup-and-delivery segment at a time, reducing the search space, whereas RHCR repeatedly replans over rolling horizons whenever the environment changes, increasing cumulative computational cost as transportation tasks increase. Despite these differences, both algorithms satisfy real-time execution requirements.

TABLE III: Comparison of average makespan.

Algorithm	8 boxes	9 boxes	10 boxes
ICBS	61.67 ± 13.87	68.20 ± 6.63	79.38 ± 15.83
RHCR	61.08 ± 11.81	67.30 ± 3.59	74.50 ± 15.91

Results are presented as mean $\pm$ standard deviation (s).

Table III shows that the makespan increased consistently with the number of boxes for both algorithms due to the greater number of transportation operations required. Nevertheless, execution times remained very similar across scenarios, with RHCR achieving a slightly lower average makespan than ICBS in the largest instances (74.50 ± 15.91 s versus 79.38 ± 15.83 s), although the overlapping standard deviations indicate comparable performance. These results suggest that RHCR's rolling-horizon replanning can adapt to environmental changes without compromising execution efficiency, while segmented ICBS provides similar efficiency through globally coordinated plans for each transportation segment and requires less cumulative planning effort. Overall, both algorithms achieved comparable execution performance.

B. Real-World Validation

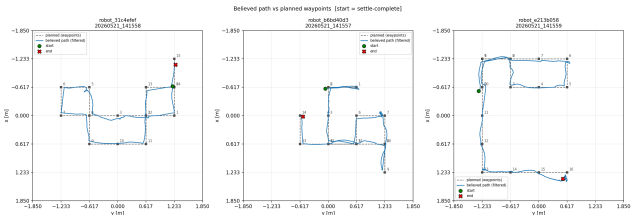
The proposed framework was deployed on the DOTS robotic platform, with the navigation speed configured to $0.1m/s$, to evaluate its feasibility under real operating conditions. Experiments were conducted using three robots and eight transportation boxes, covering the complete pipeline from task allocation and MAPF planning to navigation, pickup-and-delivery operations, and multi-robot coordination.

The experiments demonstrated that both adapted MAPF algorithms could be executed on physical robots while accounting for environmental changes caused by object manipulation. Dynamic occupancy updates allowed the planners to maintain consistent environment representations during execution, while the segmented planning strategy enabled ICBS to handle transportation stages without modifying the original planning algorithm. For RHCR, the rolling-horizon replanning process naturally incorporated workspace changes during execution.

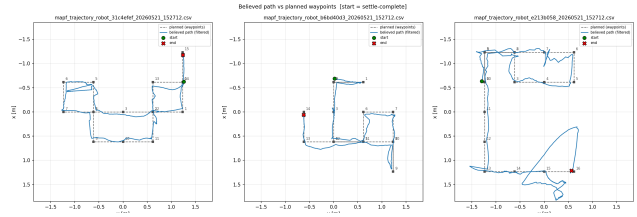
The robots coordinated within the shared workspace and completed the transportation tasks in the evaluated experiments. Although no collisions between robot bases were observed, collisions between transported boxes occurred because their dimensions were not accounted for in the MAPF planning algorithms. ROS2 integration provided communication between the planning and execution modules, allowing the complete transportation pipeline to operate on the real robotic platform. These results demonstrate the practical feasibility of applying execution-aware adaptations to MAPF algorithms in physical multi-robot systems.

However, real-world execution introduced deviations between planned and executed trajectories due to localization and motion uncertainties. Fig. 3 shows examples of accurate tracking and larger deviations, highlighting the effects of accumulated localization errors and disturbances in the robot motion system. These observations indicate that while the proposed framework enables real-world deployment, its

performance remains dependent on reliable localization and accurate low-level robot control.



(a) Accurate trajectory tracking case, where the estimated trajectory closely follows the expected path.



(b) Trajectory deviation case, showing the impact of localization and motion execution uncertainties.

Fig. 3: Expected robot trajectories (dashed lines) compared with the trajectories estimated during execution using the onboard ArUco-based localization system, illustrating the tracking deviations observed in real experiments.

The experiment logs recorded planned positions, robot poses, ArUco measurements, and odometry estimates to evaluate tracking performance. The real-world run yielded an RMSE of $0.271\mathrm{m}$, an MAE of $0.190\mathrm{m}$, a maximum error of $1.057\mathrm{m}$, and an average ArUco correction deviation of $0.133\mathrm{m}$. Robot poses were estimated by fusing wheel odometry with measurements from wall-mounted ArUco markers using an Extended Kalman Filter (EKF). Temporary occlusion of the markers by carried objects, especially near grid corners, reduced the availability of marker measurements and increased reliance on less accurate wheel odometry. The minimum inter-robot distance was $0.117\mathrm{m}$. Since only one real-world run was conducted, these results provide a preliminary assessment of system performance.

C. Discussion

The experiments demonstrate that deploying classical MAPF algorithms on real robots requires more than collision-free path planning. Reliable execution also depends on environmental consistency, communication, synchronization, and integration with the robot control system. The proposed adaptations address these challenges while preserving the original planning principles of both ICBS and RHCR. Communication delays affect the two approaches differently: ICBS requires synchronization, with replanning starting only after all robots complete the current segment, whereas RHCR replans over successive rolling windows, allowing execution to proceed without waiting for all robots to reach their waypoints.

The results also highlight the complementary characteristics of the two algorithms. ICBS provides stronger global coordination capabilities but is more affected by communication delays due to its need for synchronization. In contrast, RHCR offers greater execution flexibility, as robots can continue following their current paths during short communication interruptions. Consequently, the choice between them depends on the desired trade-off between coordination quality and flexibility.

VI. Conclusion

This paper presented an execution-aware framework to deploy classical MAPF algorithms ICBS and RHCR in autonomous multi-robot transportation systems. Instead of proposing new planning algorithms, we focused on adapting these methods to deal with the challenges of real robot execution. The proposed adaptations, including dynamic occupancy updates, segmented planning, execution synchronization, and ROS2 integration, allow both algorithms to operate on physical robotic platforms within a complete transportation pipeline.

The proposed framework was evaluated in both simulation and real experiments using the DOTS robotic platform. The results show that these execution-aware adaptations make it possible to apply adapted classical MAPF algorithms to real transportation tasks while preserving their original planning strategies. The physical experiments with three robots provided a proof-of-concept of the framework's operation on real hardware. As future work, we plan to account for the dimensions of transported boxes in trajectory planning to prevent inter-box collisions. We also plan to evaluate the framework with larger robot teams, different robot platforms, and more densely constrained environments, while exploring onboard collision avoidance and decentralized communication to improve scalability and robustness.

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